

120 YEARS OF THE SUMMER OLYMPICS

Geopolitics, Hosts, and Medal Power — 1896 to 2016

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Final Individual Project — Data Visualization, Summer 2026

28 Summer Olympiads · 146 countries & historical entities · 1,208 country-edition records

The Dataset: A Century of Medal Tallies

One row per country per Games, 1896-2016 (1916/1940/1944 absent — cancelled for the two World Wars). Every row carries: medal counts, continent, host-nation flag, historical era, and a defunct-political-entity flag (Soviet Union, East/West Germany, Czechoslovakia, Yugoslavia, and more), sourced from a public GitHub mirror and cross-referenced against the IOC's NOC code list.

28

Summer Olympiads

146

Countries & historical NOCs

6

Continents tracked

120

Years of history

Data-completeness note

The 2016 Rio source file is missing roughly 40 low-medal-count countries (a fetch-size limitation upstream) — every country of real analytical consequence is present, and any chart that depends on a complete country count is explicitly restricted or footnoted throughout this deck and the analysis notebook.

Methodology

1

Source & clean

Merged country-edition medal tallies with an NOC-to-continent lookup; hand-verified host-nation and host-city facts for all 28 editions.

2

Enrich

Derived per-edition rank, medal share, gold-conversion ratio, era buckets, and a defunct-entity flag for historical political teams.

3

Analyze

10 genuinely multi-dimensional questions — each combines at least a temporal, categorical, and quantitative dimension, not single-variable lookups.

4

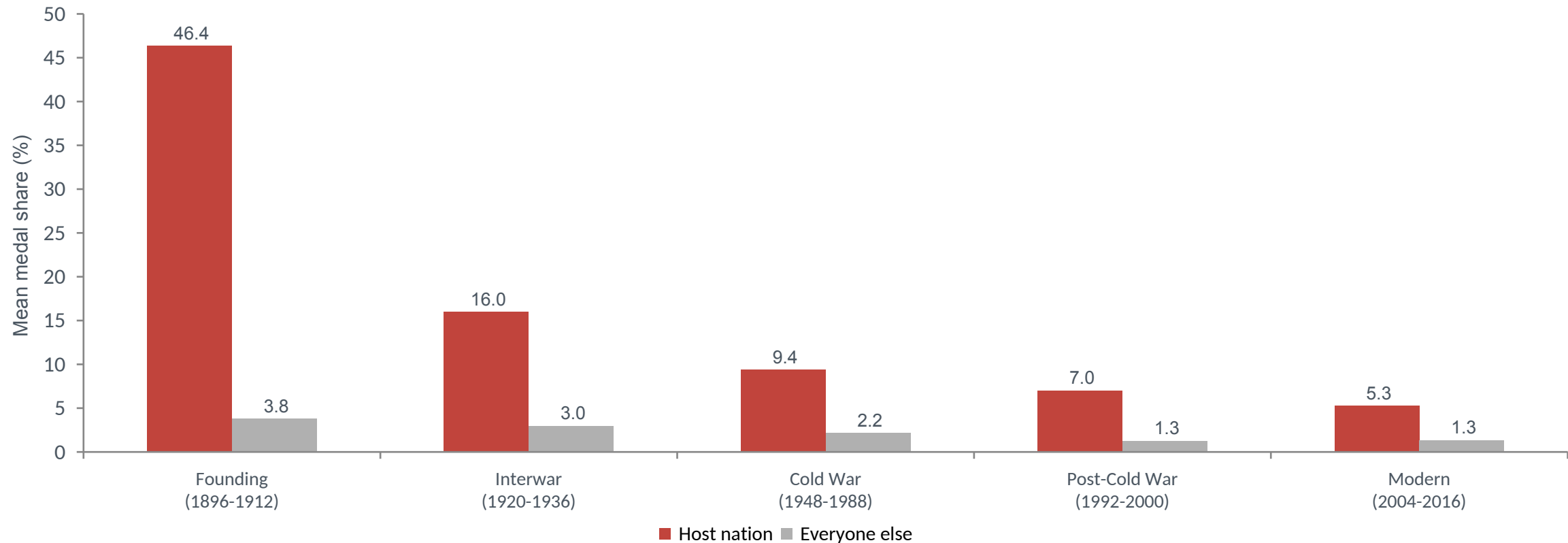
Visualize

10 explanatory Plotly visuals: colour-blind-safe (Okabe-Ito) palette, decluttered gridlines, and a stated takeaway in every chart title.

Q1

Does hosting boost a nation's medals — and has the boost shrunk over time?

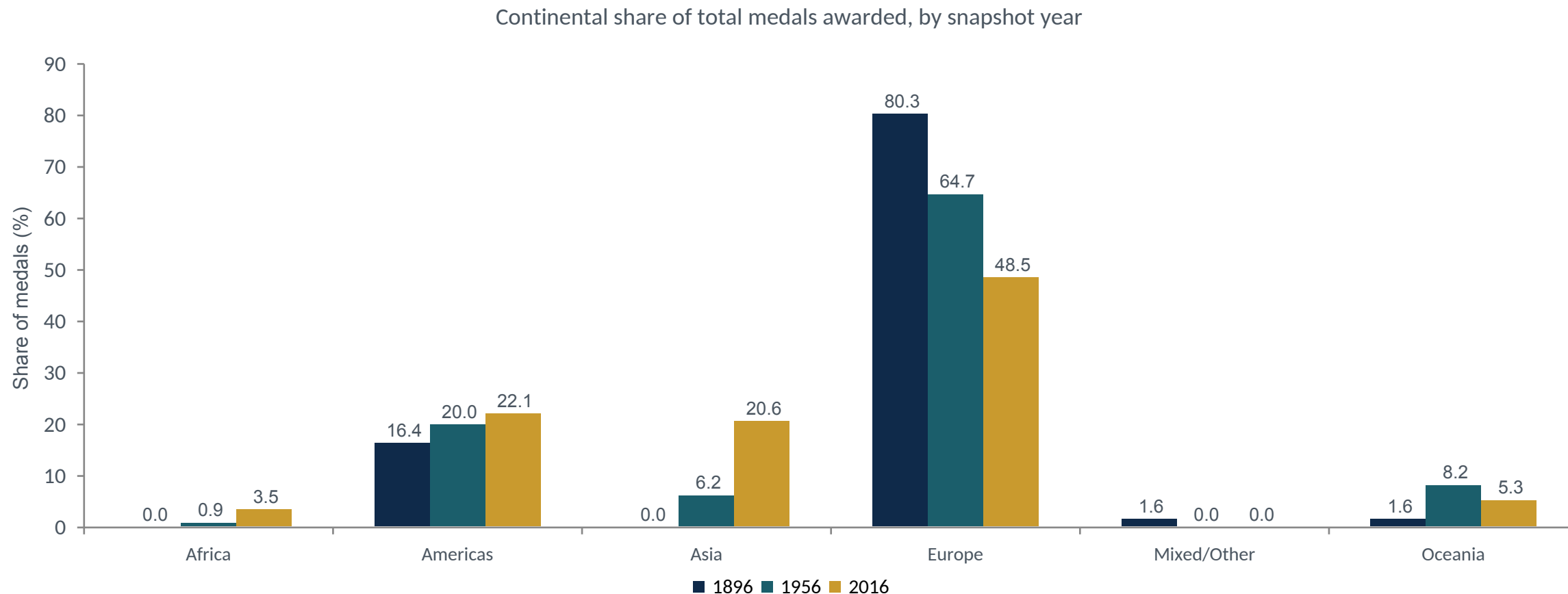
Mean share of a Games' medals: host nation vs. everyone else



Takeaway: The host boost is real in every era but has fallen from ~46% of all medals (1896-1912) to ~5% today, as fields grew from a handful of nations to 80+.

Q2

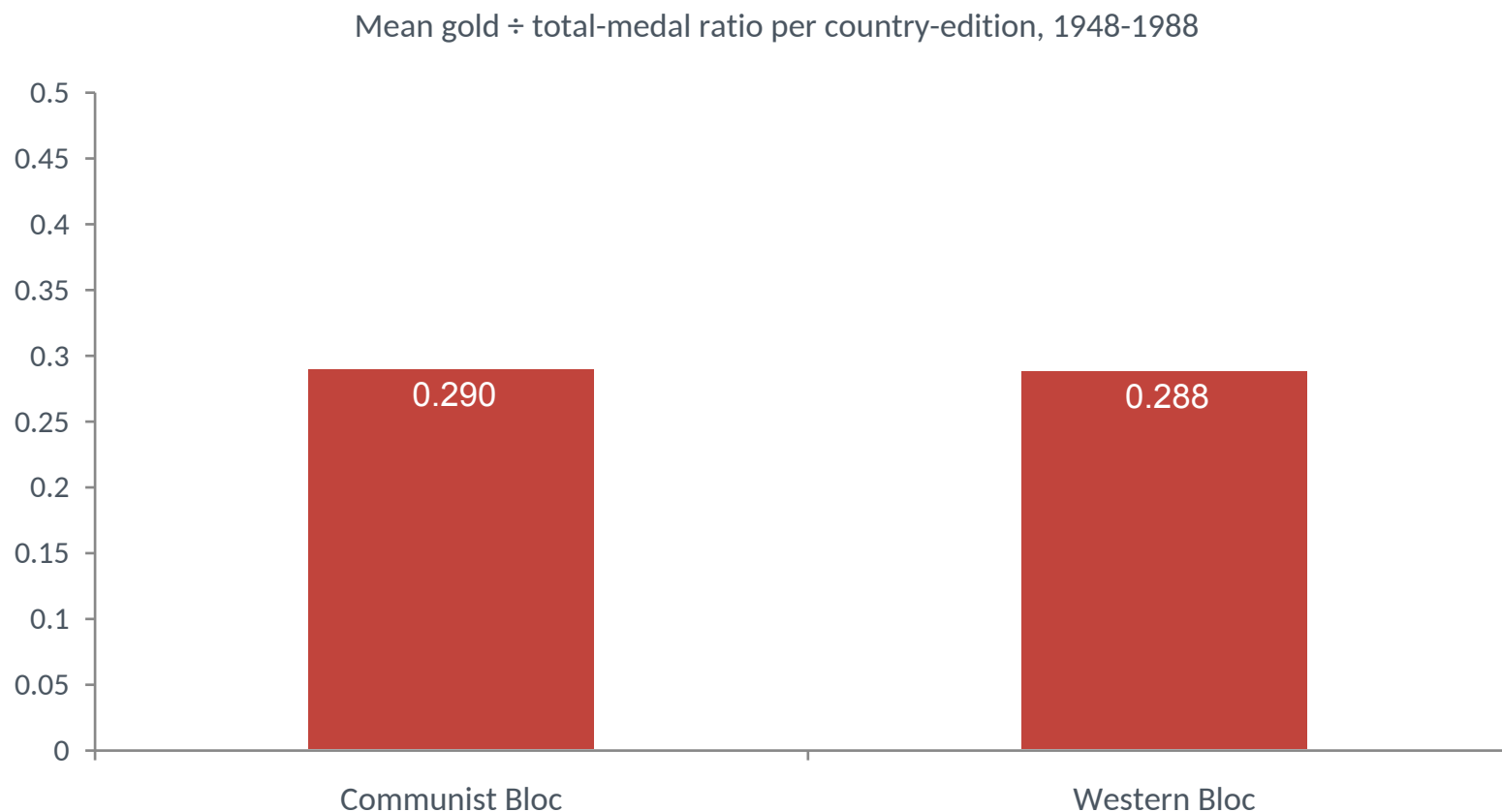
How has the continental balance of Olympic power shifted since 1896?



Takeaway: Europe's share fell from 80% (1896) to 48% (2016). The gain went overwhelmingly to Asia — near zero before the 1950s, ~21% by 2016 — not primarily the Americas.

Q3

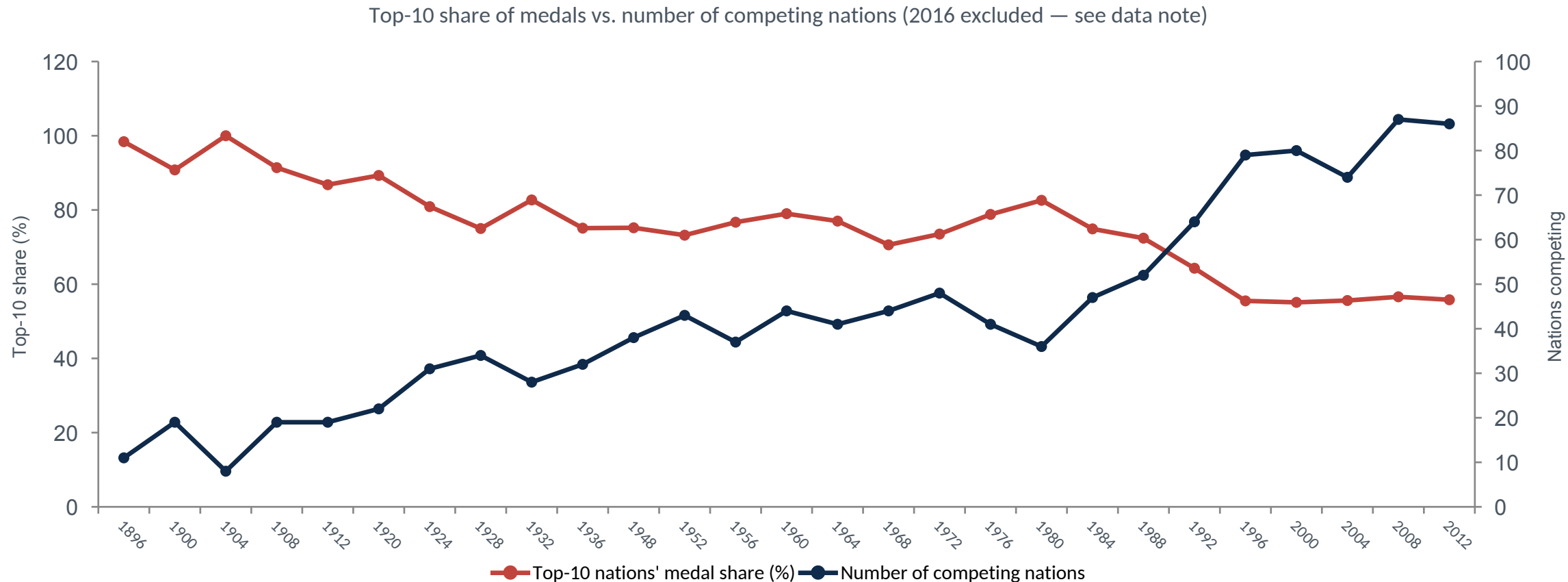
Myth-check: did the Communist Bloc convert medals to gold more efficiently?



Takeaway: The popular Cold War narrative doesn't hold: Communist Bloc (0.290) and Western Bloc (0.288) converted medals to gold at nearly identical rates. The real difference was volume, not efficiency.

Q4

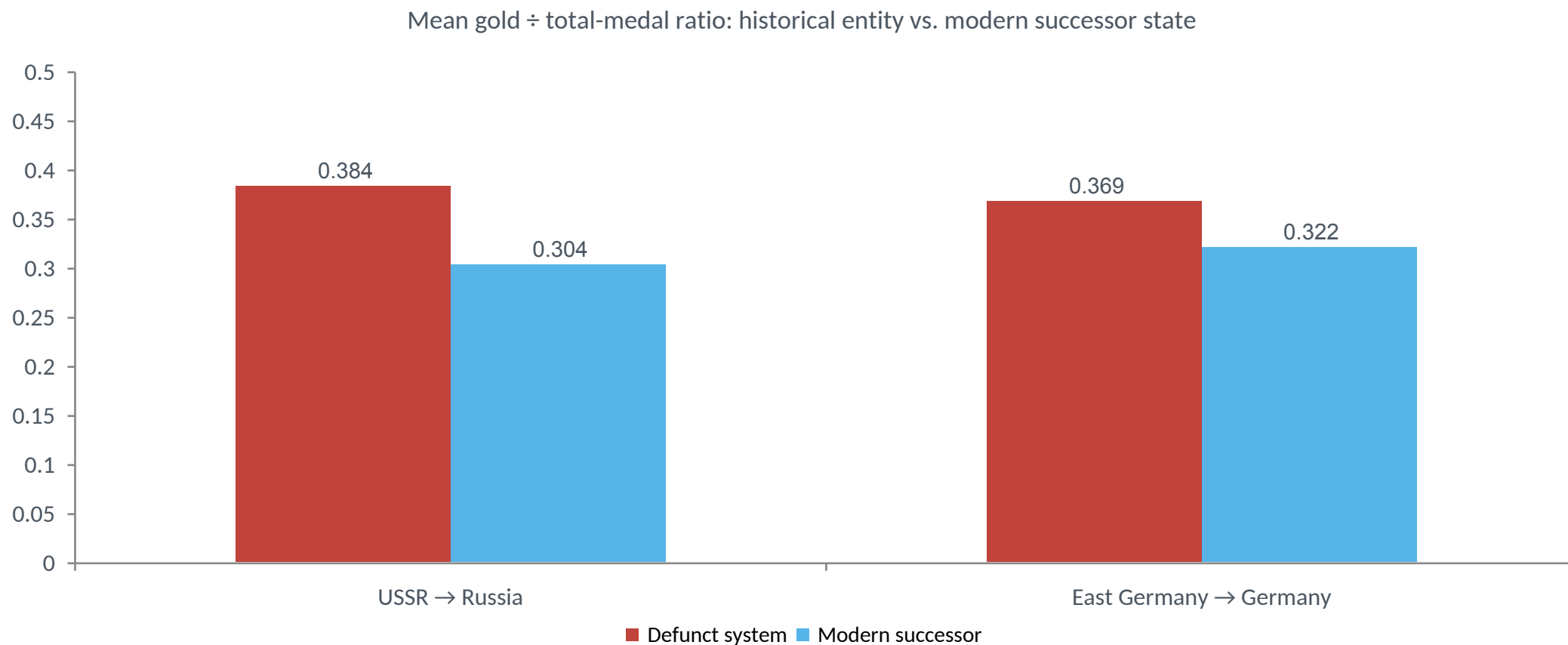
As the field grew 8x, did medal concentration fall at the same pace?



Takeaway: Concentration fell hard, from 98% (1896) to ~55% (2000s), then flattened — even in a global field of 85+ nations, the top 10 still take roughly half of all medals.

Q5

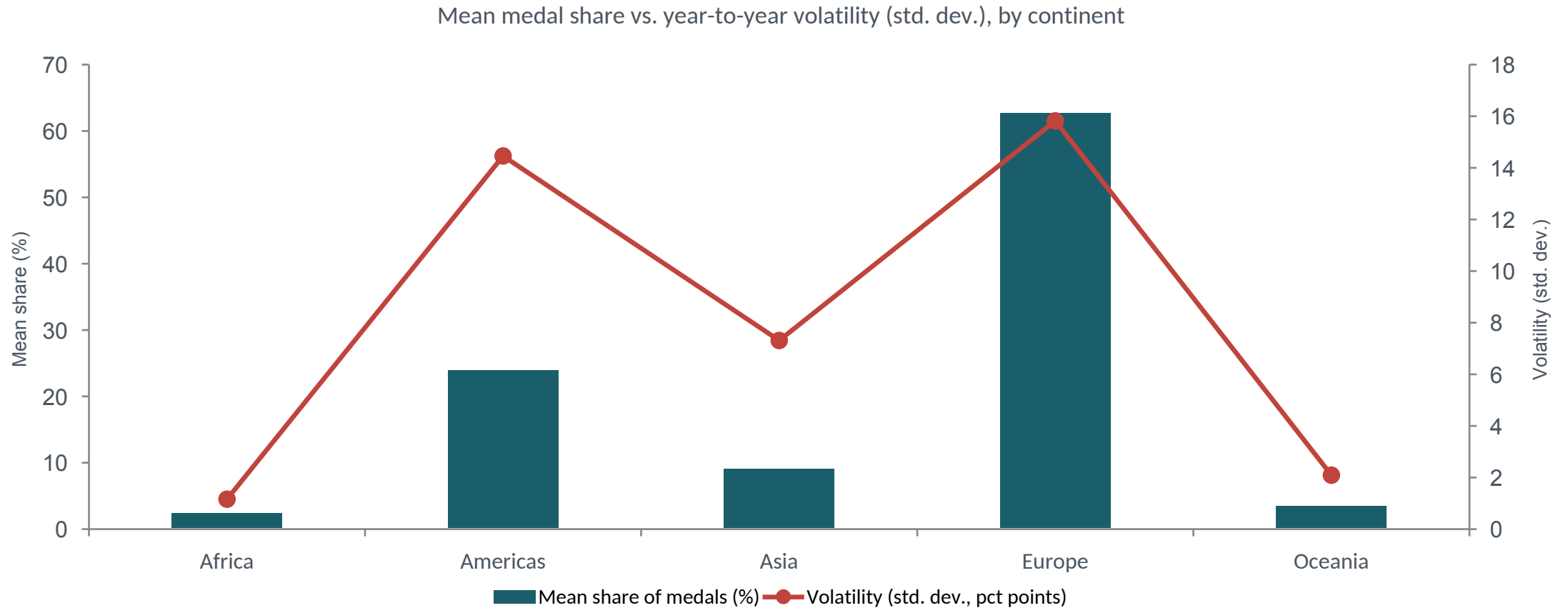
Were the defunct Cold War powerhouses more efficient than their successors?



Takeaway: Both defunct state-run systems out-converted their successors: USSR 0.384 vs Russia 0.304; East Germany 0.369 vs reunified Germany 0.322 — a real, if modest, efficiency legacy.

Q6

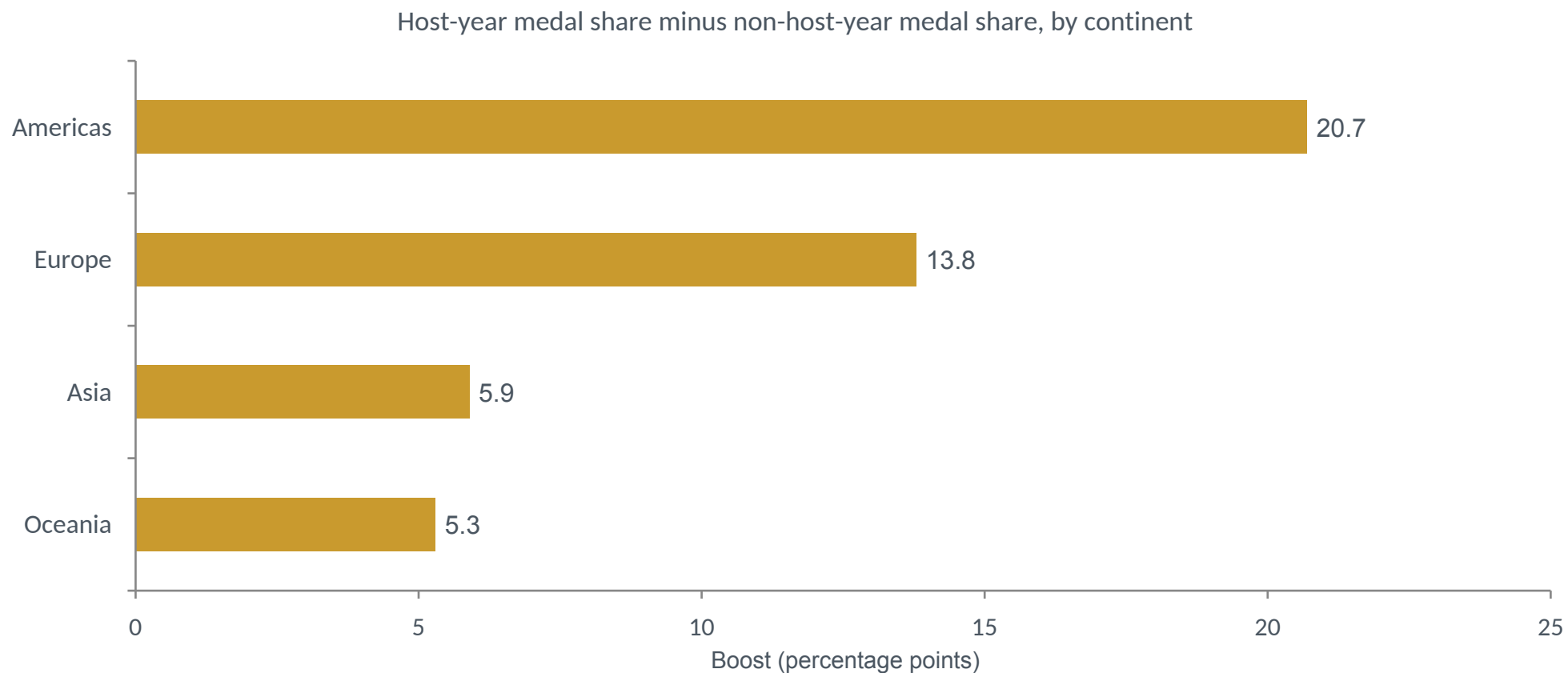
Which continent's performance is most volatile, and which most stable?



Takeaway: Europe is both the biggest (63%) and most volatile (std 15.8) continent — its share is on a long downward trend, not just noise. Africa and Oceania are small but stable.

Q7

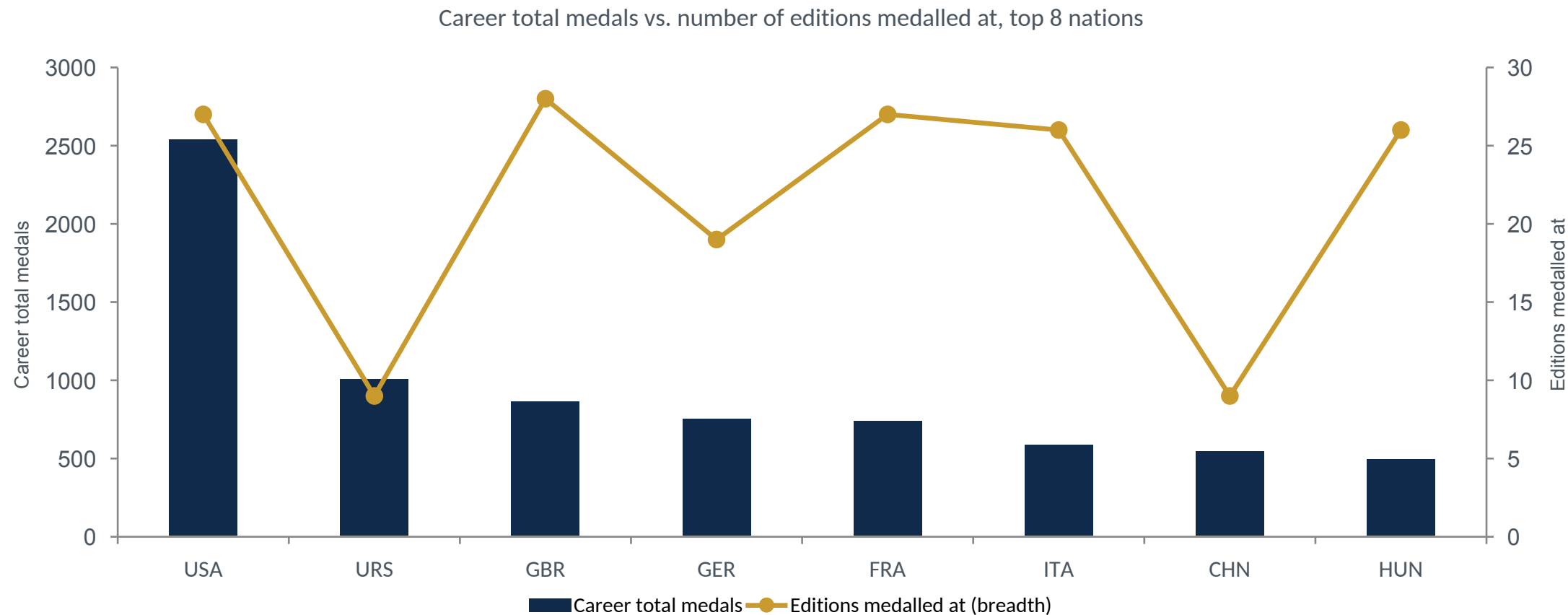
Does hosting help weaker continents more, or amplify existing strength?



Takeaway: Hosting amplifies existing strength most: the Americas (+20.7 pts) and Europe (+13.8 pts) gain far more than Asia (+5.9) or Oceania (+5.3) — it is not primarily an equaliser.

Q8

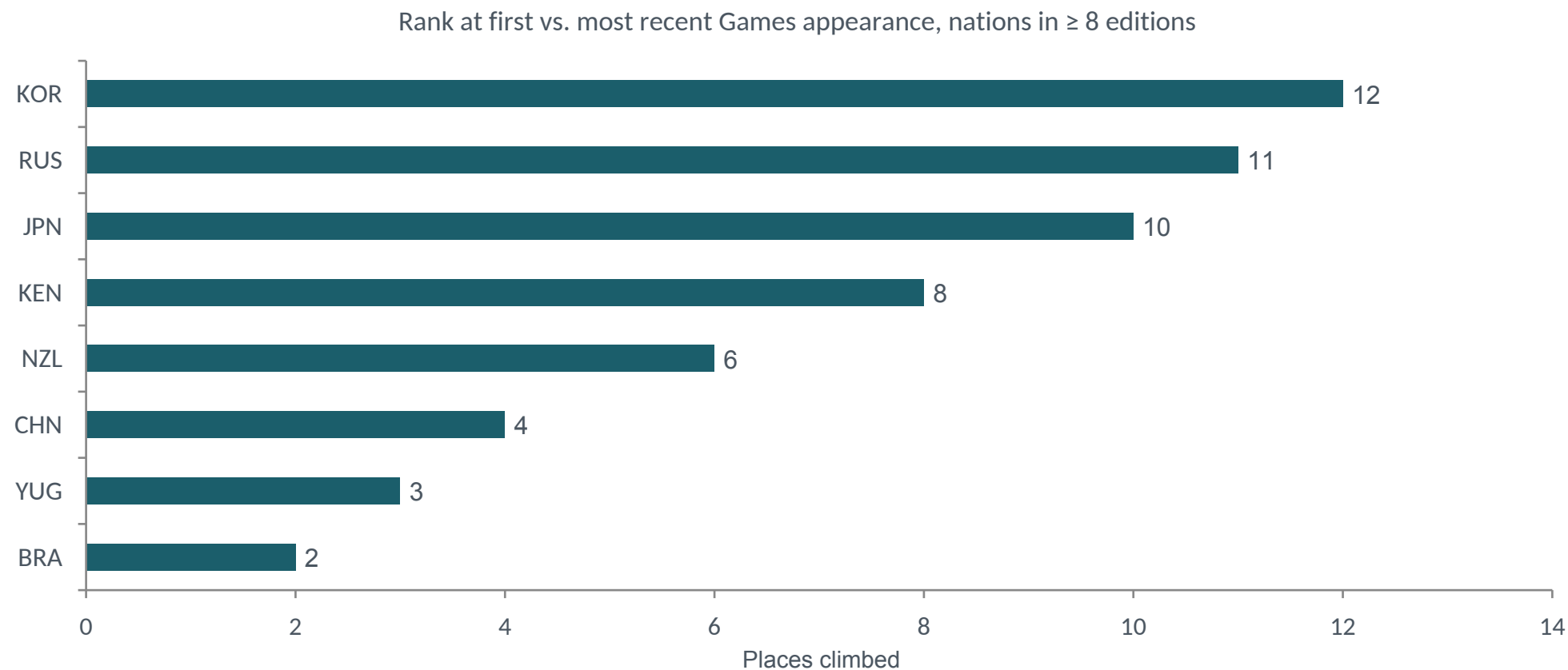
Century-long dominance, or a compressed one-era powerhouse?



Takeaway: USA and GBR dominate broadly across 27-28 editions. The USSR tells a different story: only 9 editions of existence, yet 1,010 career medals — the highest per-edition rate of any nation.

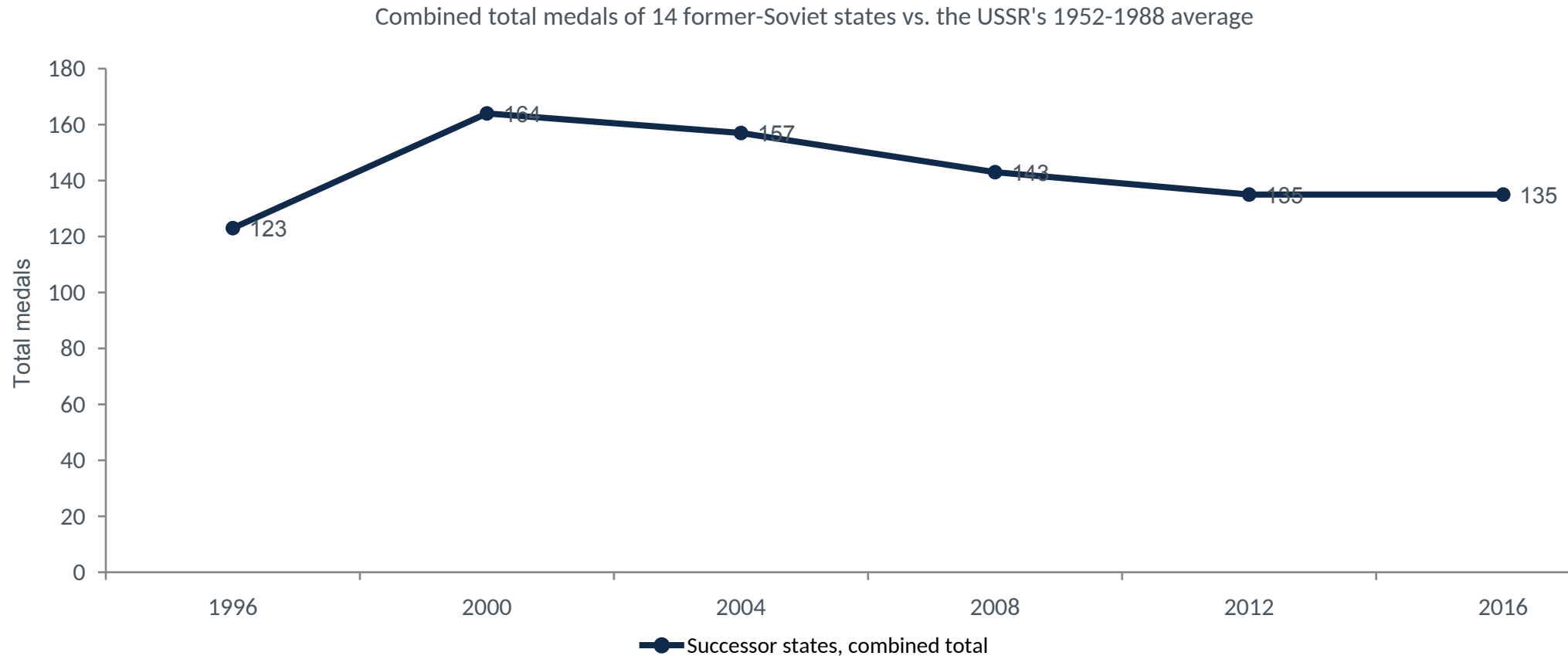
Q9

Which long-tenured nations climbed furthest in the Olympic rankings?



Takeaway: South Korea climbed 12 places (23rd to 11th), the largest rise among established nations. 3 of the top 4 climbers (KOR, JPN, KEN) are Asian or African — matching the Q2 continental shift.

Q10 Did the USSR's successor states recover its medal output, or lose ground?



Takeaway: Successor states exceeded the USSR's own historical average (112.2 medals/Games) at first (123 in 1996, peaking at 164 in 2000), then drifted back down to 135 by 2016 as the inherited talent pipeline was not fully replaced.

Interactive Dashboard

A Streamlit app (dashboard/app.py) lets a reader explore the same dataset live — filter by year range and continent, highlight any single country against its continent's average, and drive a live scatter explorer of career breadth vs. depth.

1

Overview

Choropleth map of career medal totals, top-15 leaderboard, and gold/silver/bronze split for the top 10 countries in the selected filters.

2

Continental Trends

Stacked area chart of continental medal share over time, plus a country-vs-continent-average highlight line for any country the reader picks.

3

Host & Efficiency

Host-nation boost by continent, and a live scatter explorer of career total, editions medalled, and gold-conversion ratio.

Live URL: <https://olympics-medals-final-project-rwuxvhn3cbpdntlkieyzo.streamlit.app>

GitHub: <https://github.com/Akbar40/Olympics-medals-final-project>

Deliverables & Submission



Analysis Notebook

analysis.ipynb — 10 questions, 10 Plotly visuals, CVD-safe & annotated



Dashboard

dashboard/app.py — interactive Streamlit app, deploy to Streamlit Community Cloud



Presentation

This deck, exported to PDF, with dashboard snapshots and live link



Repository

Public GitHub repo with all code, data, and this deck

Conclusions

1

Geopolitics leaves a measurable trace on sport

Cold War bloc dynamics, the Soviet collapse, and the century-long European-to-Asian power shift all show up directly in the numbers, not just the history books.

2

Hosting still helps, but scale has diluted it

The host boost is real in every era, but 1,000%-of-average effects from 1896 are gone in a globalised field — and hosting amplifies existing strength more than it lifts the weak.

3

Broader participation hasn't meant equal outcomes

Far more nations medal today than ever before, but the top 10 still take roughly half of all medals — globalisation broadened the field without levelling it.